1D Quantum Mechanics

Shooting and matching methods

Dr. Kevin Kelley, PH 385, Winter 2025

The Physics

The Schrodinger equation for stationary states in 1D

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi$$

- Wavefunction ψ must:
  - meet appropriate boundary conditions.
  - be continuous.
  - Be differentiable (except where potential makes infinite jumps)

The Physics

Bound states

- When bounded by potentials $V > E$, only certain discrete energies produce solutions to the Schrodinger equation.
- Examples:
  - Infinite square well
  - Harmonic oscillator
  - 1-D Coulomb

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These are the only 1-D bound systems with analytic solutions. All others must be solved numerically!

Method 1: Shooting

Start in the middle, satisfy BC at ends

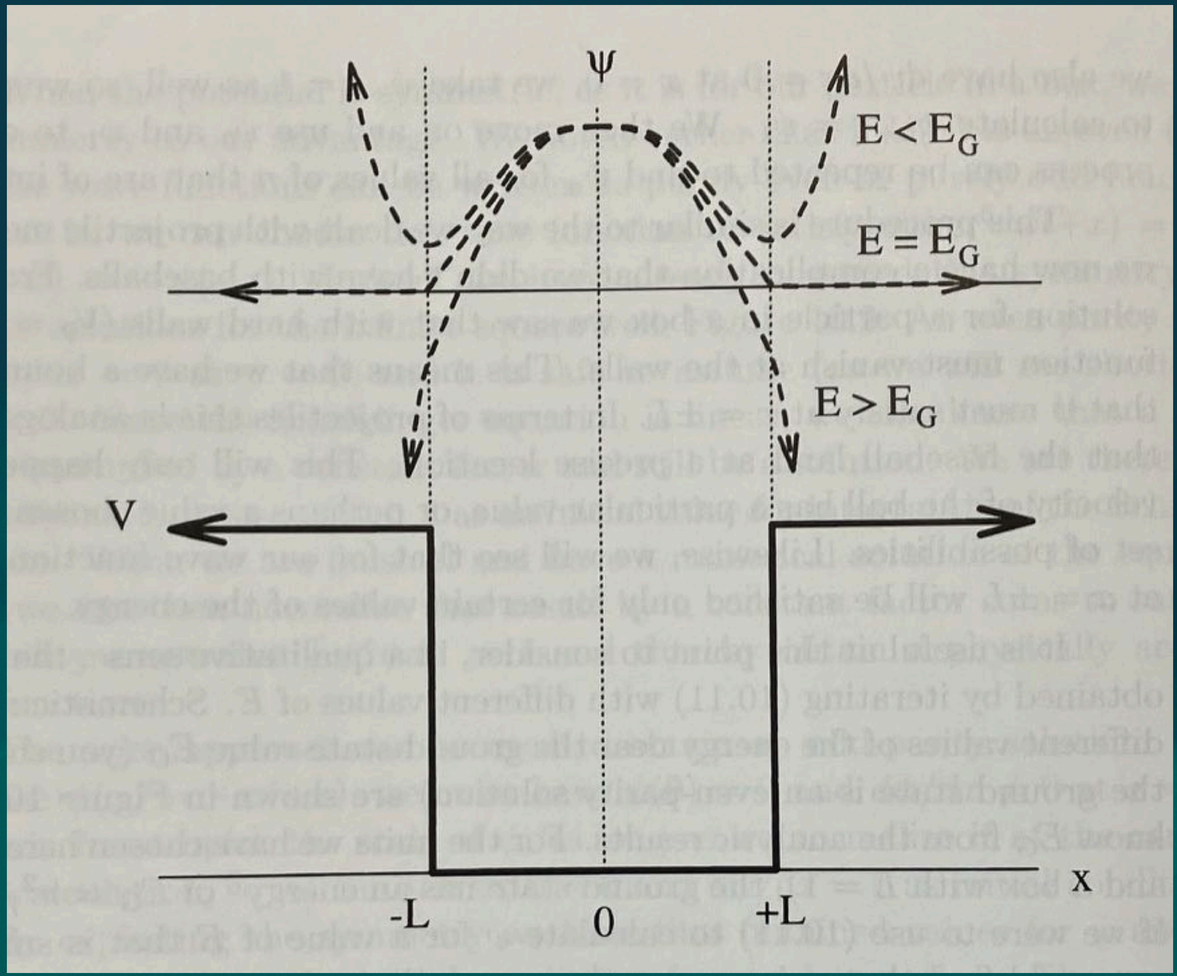
- Ideal for cases where we know where ψ has a local max/min or node.
- Iterate discretized Schrodinger equation to edges:

$$-\frac{\hbar^2}{2m}\frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{(\Delta x)^2} = \left[E - V(x_i)\right]\psi(x_i)$$

- Convenient units: $\hbar = 1, m = 1$.
- If E is too low, ψ diverges to positive values.
- If E is too high, ψ diverges to negative values.

Method 1: Shooting

Start in the middle, satisfy BC at ends



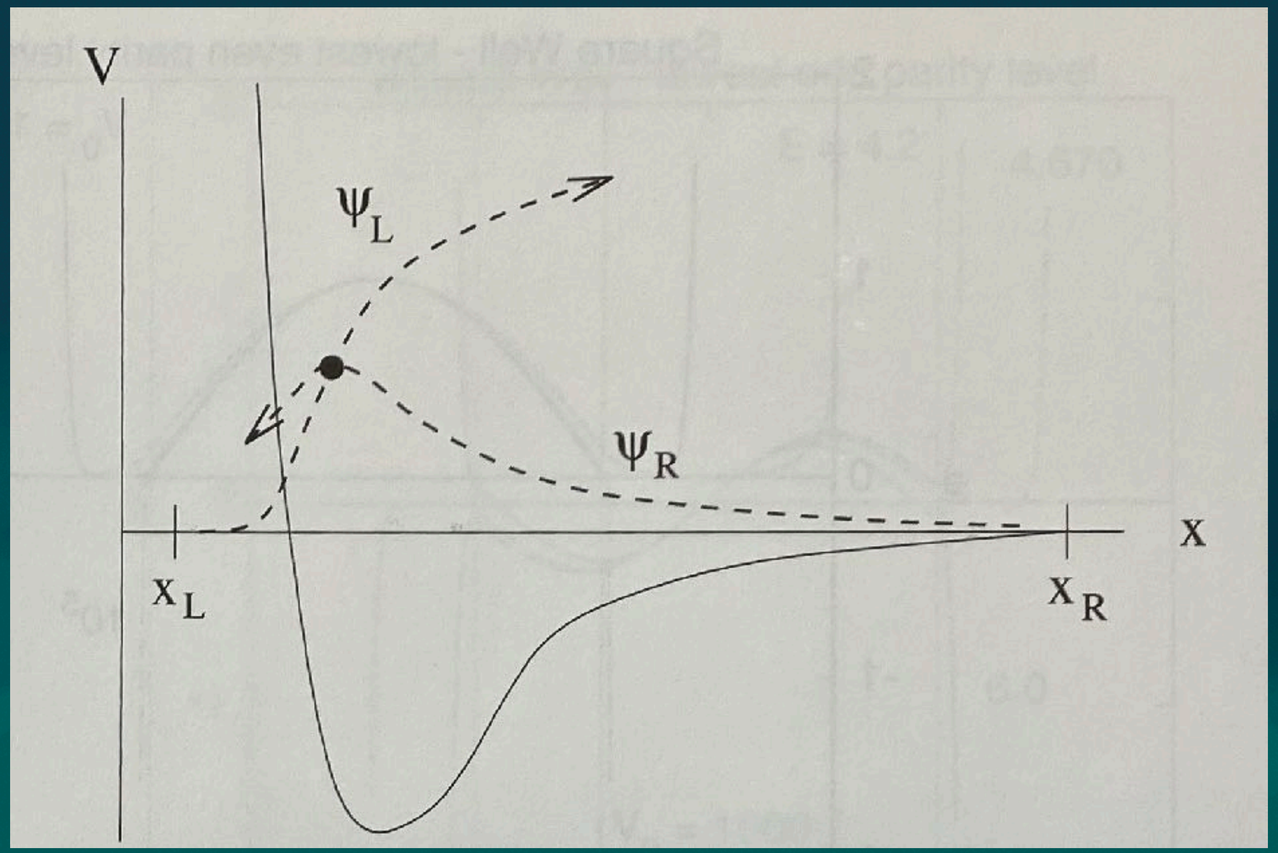
Method 2: Matching

Start at the ends, ensure continuity/differentiability

- Same discretization of the Schrodinger equation.
- Choose an obvious matching point.
- Adjust energy until the two progressions match point and slope.
- Algorithms for both methods in pseudocode file.

Method 2: Matching

Start at the ends, ensure continuity/differentiability



The Assignment

Use both methods and investigate anharmonic oscillators

- Potential $V(x) = x^n/n$ for $n = 2, 4, 6, 8, 10$ ($n = 2$ is the harmonic oscillator)
- Use both the shooting and matching methods to obtain the ground state energy and the energies of the first ten excited states.
- Plot the corresponding wavefunctions to make sure the results are reasonable!

The Assignment

Use both methods and investigate anharmonic oscillators

- Simplifications:
  - The potential is symmetric, therefore the wavefunctions must be symmetric or antisymmetric (even or odd parity).
  - Only need to model one half of the wave :)
  - Shooting: start at $x = 0$ and progress to large x.
  - Matching: start at large x and progress to $x = 0$.

The Assignment

Use both methods and investigate anharmonic oscillators

- Shooting initial and stopping conditions:
  - Initial conditions at $x = 0$:
    - Even parity: $\psi(0) = 1, \psi(-\Delta x) = 1.$
    - Odd parity: $\psi(0) = 0, \psi(-\Delta x) = -\Delta x.$
  - Stop when dE is less than some specified tolerance.

The Assignment

Use both methods and investigate anharmonic oscillators

- Matching initial and stopping conditions:
  - Initial conditions at some large value of x_l (3x width of potential at E).
    - $\psi(x_l) = 0, \psi(x_l - \Delta x) =$ (tiny).
  - Stop when dE is less than specified tolerance
    - Even parity: switch direction and reduce dE if slope at $x = 0$ changes sign.
    - Odd parity: switch direction and reduce dE if $\psi(0)$ changes sign.

The Assignment

Use both methods and investigate anharmonic oscillators

- What to turn in:
  - Your code.
  - A list of the 11 lowest energies for each of the five potentials, found using both methods.
  - A few plots showing your wavefunctions.